Deep Generative Models

Lecture 11

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AI Masters

2026, Spring

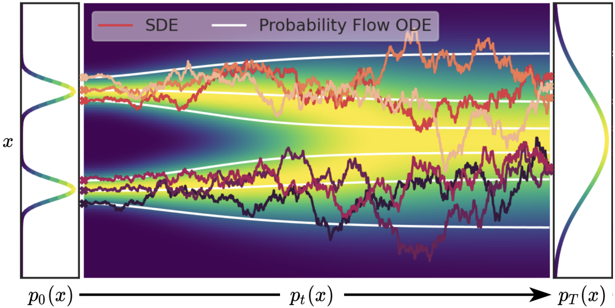
Recap of Previous Lecture

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w} \qquad \text{(SDE with the probability path } p_t(\mathbf{x}))$$

Probability Flow ODE

There exists an ODE with the identical probability path $p_t(\mathbf{x})$ of the form:

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$



Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Outline

1. Flow Matching

2. Conditional Flow Matching

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1. Flow Matching

2. Conditional Flow Matching

Continuous-Time Normalizing Flows

Let's return to ODE dynamics $\mathbf{x}_t = \mathbf{x}(t)$ in the interval $t \in [0, 1]$:

- $\mathbf{x}_0 \sim p_0(\mathbf{x})$, where $p_0(\mathbf{x})$ is a base distribution (mostly $\mathcal{N}(0, \mathbf{I})$);
- $\mathbf{x}_1 \sim p_1(\mathbf{x})$, where $p_1(\mathbf{x})$ is the true data distribution $p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity $\mathbf{v}$ determines a flow $\boldsymbol{\psi}_t$: $\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

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The velocity $\mathbf{v}$ determines a flow $\boldsymbol{\psi}_t$: $\mathbf{x}_t = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

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- It's hard to solve the continuity equation directly due to the trace term.
- There's a method (the adjoint method) that solves this equation directly, but it's unstable and unscalable.

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

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- Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- Flow matching provides scalable approach to Neural ODEs.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

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Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})\right) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

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- Flow matching provides scalable approach to Neural ODEs.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

- Approximate the true vector field $\mathbf{v}(\mathbf{x}, t)$ using $\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)$.
- Use the learned flow $\boldsymbol{\psi}_t$ for deterministic sampling: $\mathbf{x}_1 = \boldsymbol{\psi}_1(\mathbf{x}_0)$.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x}), \quad \mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$$

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Flow Matching: Training and Sampling

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Training

1. Sample a time $t \sim U[0, 1]$ and $\mathbf{x} \sim p_t(\mathbf{x})$.
2. Compute the loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Flow Matching: Training and Sampling

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Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0,\mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

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$$\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left\|\mathbf{v}(\mathbf{x},t)-\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 \to \min_{\boldsymbol{\theta}}$$

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Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
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$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Problem: The true vector field $\mathbf{v}(\mathbf{x},t)$ is **unknown**.

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

- There are infinitely many possible $\mathbf{v}(\mathbf{x}, t)$ between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- We need to select the "best" $\mathbf{v}(\mathbf{x}, t)$ and make the objective tractable.

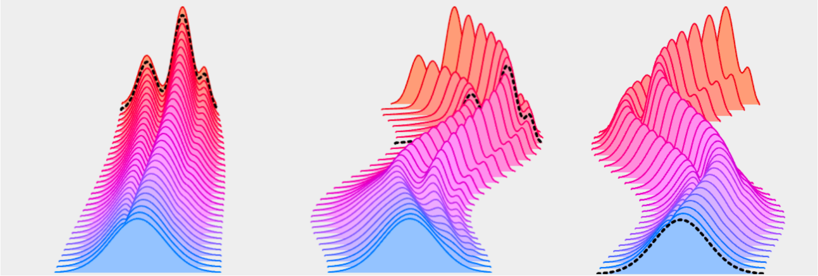
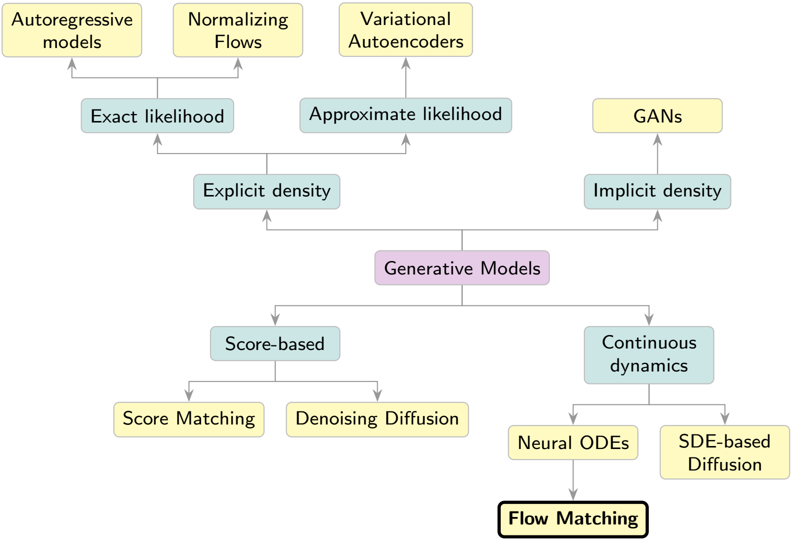


image credit: A Visual Dive into Conditional Flow Matching

Generative Models Taxonomy



Outline

1. Flow Matching

2. Conditional Flow Matching

Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**.

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Flow Matching

Let's introduce the latent variable $\mathbf{z}$:

$$p_t(\mathbf{x}) = \int p_t(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

Here, $p_t(\mathbf{x}|\mathbf{z})$ is a **conditional probability path**. The conditional probability path $p_t(\mathbf{x}|\mathbf{z})$ satisfies the KFP theorem:

$$\frac{\partial p_t(\mathbf{x}|\mathbf{z})}{\partial t} = -\operatorname{div}\left(\mathbf{v}(\mathbf{x}, \mathbf{z}, t)p_t(\mathbf{x}|\mathbf{z})\right),$$

where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

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where $\mathbf{v}(\mathbf{x}, \mathbf{z}, t)$ is a **conditional vector field**:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t) \quad \Rightarrow \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t)$$

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Flow Matching

Flow Matching (FM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

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Conditional Flow Matching (CFM)

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Theorem

If $\text{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

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Conditional Flow Matching (CFM)

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

Theorem

If $\mathrm{supp}(p_t(\mathbf{x})) = \mathbb{R}^m$, then the optimal value of the FM objective equals the optimal value of the CFM objective.

Proof

This can be proved in a similar way as in the denoising score matching theorem.

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Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \left\| \mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 =$$
$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2$$

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Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

$$= \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left\|\mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2$$

Proof

$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] + \text{const}(\boldsymbol{\theta})$$

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

Conditional Flow Matching

Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

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$$\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 =$$

$$= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x}|\mathbf{z})} \left[\left\|\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right\|^2 - 2\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] + \text{const}(\boldsymbol{\theta})$$

$$\mathbb{E}_{p_t(\mathbf{x})}\left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] = \int p_t(\mathbf{x}) \left[\mathbf{v}_{\boldsymbol{\theta}}^{\top}(\mathbf{x}, t) \left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x}, \mathbf{z}, t) d\mathbf{z}\right)\right] d\mathbf{x}$$

Tong A., et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023

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$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 =$$
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$$\mathbb{E}_{p_t(\mathbf{x})}\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x},t)\mathbf{v}(\mathbf{x},t)\right] = \int p_t(\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x},t)\left(\int p_t(\mathbf{z}|\mathbf{x})\mathbf{v}(\mathbf{x},\mathbf{z},t)d\mathbf{z}\right)\right]d\mathbf{x} =$$
$$= \int\int p_t(\mathbf{x})p_t(\mathbf{z}|\mathbf{x})\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x},t)\mathbf{v}(\mathbf{x},\mathbf{z},t)\right]d\mathbf{z}d\mathbf{x}$$

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Theorem

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})} \left\|\mathbf{v}(\mathbf{x},t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x},t)\right\|^2 =$$

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$$= \mathbb{E}_{p(\mathbf{z})}\mathbb{E}_{p_t(\mathbf{x}|\mathbf{z})}\left[\mathbf{v}_{\boldsymbol{\theta}}^\top(\mathbf{x},t)\mathbf{v}(\mathbf{x},\mathbf{z},t)\right]$$

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Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

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Training

1. Sample a time $t \sim U[0, 1]$ and $\mathbf{z} \sim p(\mathbf{z})$.
2. Draw $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
3. Compute the loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

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Conditional Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}(\mathbf{x}, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) \right\|^2 \rightarrow \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p(\mathbf{z})} p_0(\mathbf{x}|\mathbf{z}) = \mathcal{N}(0, \mathbf{I}); \quad \mathbb{E}_{p(\mathbf{z})} p_1(\mathbf{x}|\mathbf{z}) = p_{\text{data}}(\mathbf{x}).$$

Training

1. Sample a time $t \sim U[0, 1]$ and $\mathbf{z} \sim p(\mathbf{z})$.
2. Draw $\mathbf{x}_t \sim p_t(\mathbf{x}|\mathbf{z})$.
3. Compute the loss $\mathcal{L} = \left\| \mathbf{v}(\mathbf{x}_t, \mathbf{z}, t) - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2$.

Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0 = 0, t_1 = 1).$$

Lipman Y., et al. Flow Matching for Generative Modeling, 2022

Summary

- SDEs can be reversed using the score function.
- It's possible to train continuous-in-time score-based generative models using forward and reverse SDEs.
- Discretizing the reverse SDE yields ancestral sampling of the DDPM.
- Flow matching suggests fitting the vector field directly.
- Conditional flow matching introduces the latent variable $\mathbf{z}$, reformulating the initial task in terms of conditional dynamics.